

25/12/23

Mock Test Subjective.

Physics.

1. (3) concave mirror

2. $u = -10\text{cm}$

$m = -3$

$-3 = \frac{v}{-10}$

$\bullet 30\text{cm} = v$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{30} + \frac{1}{10}$$

$$= \frac{1+3}{30}$$

$$= \frac{4}{30}$$

$$f = \frac{30}{4} = 7.5\text{cm}$$

(3) 30cm, 7.5cm

3. (4) Both 2 and 3.

4(i) Principal focus of spherical mirror—

①

When light rays ~~go~~ coming from infinity parallel to principal axis after reflection ~~converges~~ or appears to converge at a point behind the mirror is called principal focus.

(ii) Centre of curvature of spherical mirror—
The centre of the sphere from which the spherical mirror was a part of is known as centre of curvature.

5. Given (Convex lens)

(i) $m = \frac{-1}{5}$

$$v = +18 \text{ cm}$$

$$m = \frac{v}{u}$$

$$\frac{-1}{5} = \frac{18}{u}$$

$$u = \frac{18 \times 5}{-1}$$

$$= -90 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$= \frac{1}{18} + \frac{1}{90}$$

(2)

$$\frac{1}{f} = \frac{5+1}{90}$$

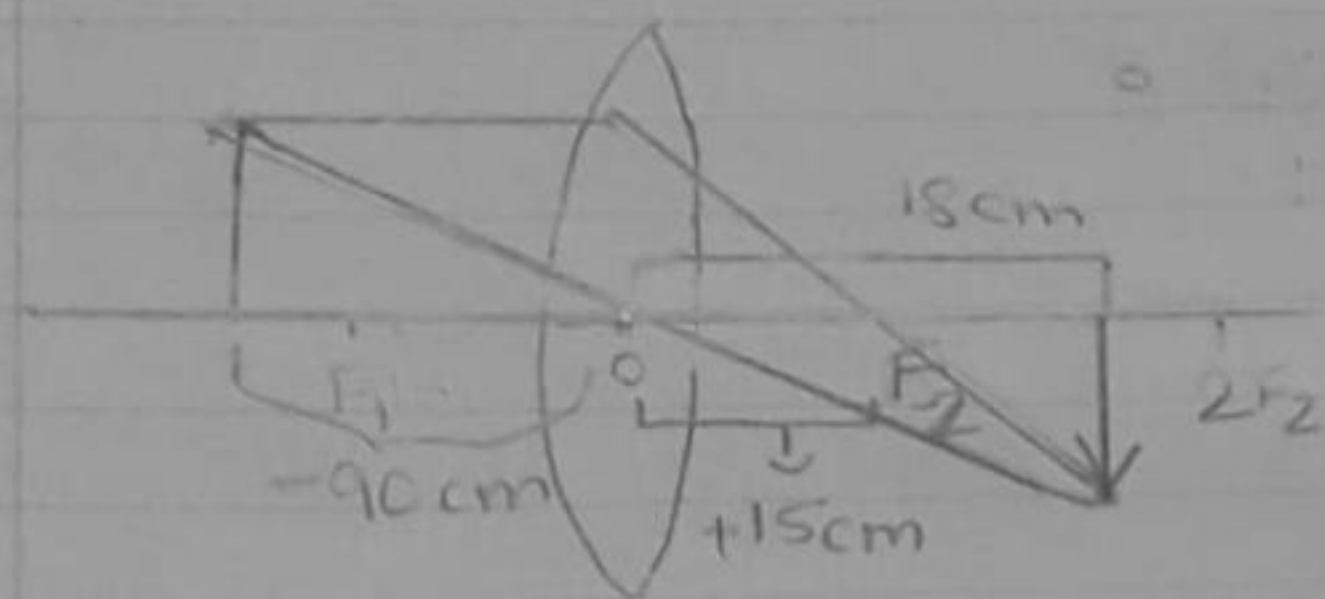
$$= \frac{6}{90}$$

$$f = \frac{90}{6} = +15\text{cm}$$

$$\begin{array}{r} 15 \\ 6 \overline{) 90} \\ \underline{30} \end{array}$$

(ii) Nature: The lens is convex lens.
Convex lens is convergent.

(iii)



6. (i) Given (Concave mirror)

$$h_o = +7\text{cm}$$

$$u = -27\text{cm}$$

$$f = -18\text{cm}$$

$$v = ?$$

$$h_i = ?$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{-1}{18} = \frac{1}{v} - \frac{1}{27}$$

$$\frac{1}{v} = \frac{1}{27} - \frac{1}{18}$$

$$= \frac{2-3}{54}$$

$$= \frac{-1}{54}$$

$$v = -54 \text{ cm}$$

$$m = \frac{-v}{u} = \frac{h_i}{h_o}$$

$$= \frac{-(-54)}{+27} = 2$$

$$m = -2$$

$$-2 = \frac{h_i}{+7}$$

$$-14 \text{ cm} = h_i$$

Size: The image formed is 14 cm from the principal axis.

Nature: The image formed is real and inverted. The image formed is twice the size of the object.

(ii) Given (convex mirror)

$$2f = +5m$$

$$u = -20m$$

$$v = ?$$

find

$$2f = +5m$$
$$f = \frac{+5}{2}m$$

$$2f = +5m$$
$$f = \frac{+5}{2}m$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$\frac{2}{5} = \frac{1}{v} + \frac{1}{20}$$

$$\frac{2}{5} = \frac{1}{v} + \frac{1}{20}$$

$$\frac{1}{v} = \frac{2}{5} + \frac{1}{20}$$

$$\frac{2}{5} + \frac{1}{20} = \frac{1}{v}$$

$$= \frac{9}{20}$$

$$\frac{1}{v} = \frac{8+1}{20}$$

$$v = \frac{20}{9}$$

$$= \frac{9}{20}$$

$$= +2.22m$$

$$v = +\frac{20}{9}m$$

$$m = \frac{-v}{u}$$

$$= \frac{-\frac{20}{9}}{-20}$$

$$= \frac{-20}{9} \times \frac{1}{(-20)} = \frac{1}{9}$$

(5)

Position: The image is formed $+2.22\text{m}$ behind the mirror.

Nature: The image formed is virtual and erect. The nature of the lens is diverging.

Size: The image formed is $\frac{1}{9}$ th the size of the object.

(iii) a) The nature of this mirror is real and inverted. The mirror is converging.

b) This Concave mirror is used so that dentists get an enlarged image of the tooth.